StockEvent Lens — Batch 020 Intelligence Dashboard

Articles 096–100 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_20 (Articles 096-100)`

TEST 096
Original — 100% Base
Big Tech AI capex faces grid bottleneck; power demand projected to surge 1,100%+ to 315 GW by 2033[cite: 14].

TEST 097
Repeat — 95% Overlap
Restates 1,100% power surge to 315 GW globally (200 GW US) with zero new data[cite: 14].

TEST 098
Update — +20% New Info
Ratepayer Protection Pledge signatories release new quarterly implementation timetable[cite: 14].

TEST 099
Conflict — Dispute (150 GW)
Contradicts energy projections: claims global AI demand will only reach 150 GW (+500%)[cite: 14].

TEST 100
Different — 0% Overlap
NVIDIA ETF concentration risks and Michael Burry 'model collapse' warning starting Group 040[cite: 14].

Technical Analysis: Dense Vector Blindness vs. Temporal Contradiction Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *numerical accuracy or forecast fidelity*. Divergent energy growth figures (e.g., +1,100% to 315 GW in test_096 vs +500% to 150 GW in test_099) yield extremely high vector similarity (~0.88) because both texts share identical domain tokens ("AI data centers", "electricity demand", "grid capacity", "gigawatts").

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard numeric or forecasted gigawatt growth dispute.
```

1. Dense Embedding Blindness
Dense embeddings project words into high-dimensional space where "315 gigawatts (+1,100%)" and "150 gigawatts (+500%)" share near-identical contextual embeddings. Consequently, standard vector cosine metrics fail to recognize the severe numerical contradiction.

2. LLM Evaluation Override
To ensure forecasting fidelity, the LLM reasoning layer performs structured attribute extraction and numeric threshold validation, overriding vector similarity scores when $\Delta F_{contradiction} = 1.0$ is established.

Article Relationship & Percentage Breakdown (Batch 020)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_096_original`	Original (%5=1)	AI power demand, Kobeissi Letter, 315 GW global, $1.3T capex, Ratepayer Pledge[cite: 9, 14]	100%	Baseline	Establishes breaking story frame on hyperscaler power bottlenecks and 1,100%+ power growth projections[cite: 14].
`test_097_repeat`	Repeat (%5=2)	AI chip electricity, 1,100% surge, 315 gigawatts, 200 GW US, grid capacity[cite: 10, 14]	95%	0% New Facts	Restates identical power growth metrics, global gigawatt targets, and grid constraints without new facts[cite: 14].
`test_098_update`	Update (%5=3)	**Quarterly implementation timetable**, grid-upgrade cost reporting, Ratepayer Pledge[cite: 11, 14]	83%	+20% Policy Facts	Advances event state by announcing a quarterly implementation and reporting timetable for pledge signatories[cite: 14].
`test_099_conflict`	Conflict (%5=4)	**150 gigawatts dispute**, 500% surge claim, contradictory market forecast[cite: 12, 14]	88% Match	$\Delta F_{contradiction} = 1.0$	Directly disputes baseline forecast, claiming global AI power demand caps at 150 GW (+500%)[cite: 14].
`test_100_different`	Different (%5=0)	NVIDIA $5.4T cap, VOO & QQQ ETF weighting, Michael Burry, model collapse[cite: 8, 14]	0%	Group 040 Trigger	Zero domain overlap; triggers standalone Group 040 event cluster on ETF concentration and AGI warnings[cite: 14].

Group 039 — Consolidated Event Summary

4 Articles (096–099)

Big Tech faces critical grid bottlenecks as hyperscaler AI capex approaches $1.3 trillion in 2027, with companies expanding energy infrastructure through nuclear upgrades and backup generator deals (Articles 096–097)[cite: 9, 10, 14]. Corporate compliance state advanced as signatories of the Ratepayer Protection Pledge—including Amazon, Google, Meta, Microsoft, and Oracle— released a new quarterly implementation timetable requiring initial progress reports next quarter (Article 098)[cite: 11, 14]. However, long-term power demand projections are in direct conflict: initial reports projected 1,100%+ power growth to 315 GW globally by 2033 (200 GW in the US), whereas a contradictory report claims global AI power demand will only reach 150 GW by 2030 (a 500% increase) (Article 099 ; $\Delta F_{contradiction} = 1.0$)[cite: 12, 14].

✓ Update Preserved: Quarterly Implementation Timetable Issued
✓ Conflict Preserved: 315 GW (+1,100%) vs 150 GW (+500%) Demand Forecast
✓ Duplicate Removed: Article 097 Consolidated

Group 040 — Standalone Summary

1 Article (100)

NVIDIA's $5.4 trillion market capitalization creates heavy portfolio concentration in broad-market ETFs such as Vanguard S&P 500 ETF (VOO at 8.08%) and Invesco QQQ Trust (QQQ at 8.2%)[cite: 8, 14]. Concurrently, investor Michael Burry warned that recursive training on synthetic AI data risks triggering 'model collapse' and hindering AGI progress, raising questions about mega-cap valuations (Article 100) [cite: 8, 14].

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 039

Batch 020 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

test_096_original.json — Group 039

```
{
  "id": 61991451,
  "headline": "Big Tech’s Data Center Boom Could Be St
  "symbols": ["AMZN", "GNRC", "GOOGL", "META", "MSFT",
  "summary": "Big Tech's AI expansion faces a power bo
}
```

Establishes 315 GW (+1,100%) baseline.

test_098_update.json — Group 039

```
{
  "id": 990000098,
  "headline": "Tech giants sign Ratepayer Protection P
  "symbols": ["AMZN", "GOOGL", "META", "MSFT", "NVDA"]
  "summary": "Hyperscalers committed to funding energy
}
```

Adds quarterly reporting timetable.

test_099_conflict.json — Group 039

```
{
  "id": 990000099,
  "headline": "Report claims AI electricity demand wil
  "symbols": ["NVDA"],
  "summary": "A conflicting market study estimates glo
}
```

Disputes 315 GW (+1,100%) vs 150 GW (+500%).